

Tangents and normals



- 1 Find the gradient of the normal to the curve $y = x^2 + \cos(\pi x)$, at the point where $x = 1$.
- 2 For the following functions, find the equation of the tangent at the given point:
 - a $y = -5x \sin x$ at $(\pi, 0)$.
 - b $y = \cos 6x$ at the point where $x = \frac{\pi}{18}$.
- 3 Consider the function $f(x) = \cos 2x - \sqrt{3} \sin 2x$.
 - a Find the first positive value of x at which the curve intersects the x -axis.
 - b Find the gradient of the tangent at this point.
 - c Hence determine the positive angle, θ , that the tangent makes with the positive x -axis at this point. Round your answer to the nearest degree.
- 4 Consider the function $f(x) = -4 \cos 3x$.
Find θ , the angle that the tangent at $x = \frac{\pi}{12}$ makes with the positive x -axis. Round your answer to the nearest degree.
- 5 For each of the following functions, find:
 - i The derivative function.
 - ii The gradient of the normal at the point $\left(\frac{\pi}{4}, 1\right)$.
 - iii The equation of the normal at the point $\left(\frac{\pi}{4}, 1\right)$.
 - a $y = \sin 2x$
 - b $y = \sqrt{2} \sin x$
 - c $y = \sqrt{2} \cos x$
- 6 For the function $y = \frac{\sin x}{x}$, find:
 - a The derivative function.
 - b The gradient of the normal at the point $\left(\frac{\pi}{2}, \frac{2}{\pi}\right)$.
 - c The equation of the normal at the point $\left(\frac{\pi}{2}, \frac{2}{\pi}\right)$.



Trigonometric graphs

- 7 Consider the function $y = \frac{\sin x}{1 + \cos x}$.
- Find $\frac{dy}{dx}$.
 - Determine the number of turning points function y has.
- 8 Find the x -coordinate of each point of inflection on the curve $y = 3 \cos \left(2x + \frac{\pi}{4} \right)$ for the interval $0 \leq x \leq 2\pi$.
- 9 Consider the function $f(x) = -4 \sin \left(x + \frac{\pi}{6} \right)$ on the interval $0 \leq x \leq 2\pi$.
- Find the coordinates of any turning points.
 - Find an expression for $f''(x)$.
 - Hence classify the turning points.
- 10 For each of the following functions on the interval $0 \leq x \leq 2\pi$:
- Find the axes intercepts.
 - Find an expression for $f'(x)$.
 - Determine the coordinates of any stationary points.
 - Find an expression for $f''(x)$.
 - Hence classify the stationary points.
 - Sketch the graph of $f(x)$.
- a $f(x) = 2 \sin x$ b $f(x) = \cos 3x$ c $f(x) = x \cos x$ d $f(x) = e^{-x} \cos x$
e $f(x) = \sin^2(x)$ f $f(x) = \cos^3(x)$ g $f(x) = x + \sin x$

Applications

- 11 Determine the possible values of k , given that $y = \cos kx$ satisfies the following equation for all values of x :

$$y'' + 25y = 0$$

- 12** The displacement of an object is a measure of how far it is from a fixed origin point at time, t seconds. Displacement to the left of the origin point is negative and displacement to the right of the origin is positive. At time t seconds, velocity is given by $v(t) = x'(t)$, and acceleration is given by $a(t) = x''(t)$.



If the displacement of an object x about a fixed origin is given by:

$$x(t) = -5 \cos 2t$$

- a** Describe the initial position of the object, from the fixed origin point.
 - b** Write an expression for $v(t)$.
 - c** Show that $a(t) = -4x(t)$.
 - d** Find the acceleration of the object, when $t = \frac{\pi}{2}$.
- 13** A particle moves in a straight line and its displacement, x cm, from a fixed origin point after t seconds is given by the function:

$$x(t) = \sin t - \sin t \cos t - 5t$$

- a** State the initial displacement of the particle.
 - b** The velocity is the rate at which displacement changes over time. Find an equation for $v(t)$, in terms of $\cos t$.
 - c** Hence find the initial velocity, $v(0)$.
- 14** The arm of a pendulum swings between its two extreme points A to the left and B to the right. Its horizontal displacement x cm from the centre of the swing, at time t seconds after it starts swinging, is given by:

$$x(t) = 16 \sin 3\pi t$$

- a** Find the initial position of the pendulum.
- b** Find the maximum distance of the pendulum from the central position of its swing.
- c** Find an expression for the velocity function, $v(t) = x'(t)$.
- d** Find the first two times at which the pendulum comes to rest, $v(t) = 0$.
- e** Find the displacement of the pendulum when it first comes to rest.
- f** Find the displacement of the pendulum when it comes to rest for the second time.
- g** Hence determine the distance between the two points A and B .

- 15 The arm of a pendulum swings between its extreme points A to the left and B to the right. Its horizontal displacement x cm from the centre of the swing at time t seconds after it starts swinging is given by:

$$x(t) = 19 \sin 4\pi t$$

- a Find an expression for the velocity function, $v(t) = x'(t)$.
- b Find the maximum velocity of the pendulum.
- c Find the first two times, after it starts swinging, at which the pendulum reaches its maximum velocity, $v'(t) = 0$ and $x'(t) < 0$.
- d Describe the position of the pendulum when it reaches its maximum velocity.